House Price Prediction using Machine Learning - Anshika Bhatt

Objective

The objective of this project was to build *machine learning models capable of predicting house prices based on various housing characteristics* such as area, number of bedrooms, bathrooms, stories, parking facilities, furnishing status, and other amenities. The project also aimed to *identify the factors that most strongly influence house prices.*

Dataset

The **Housing Prices Dataset** (https://www.kaggle.com/datasets/yasserh/housing-prices-dataset) from Kaggle was used for this analysis. The dataset *contains 545 housing records* with multiple numerical and categorical features related to residential properties.

Models Used

Two regression models were implemented:

- Linear Regression
- Random Forest Regressor

The dataset was preprocessed using one-hot encoding for categorical variables and divided into training and testing sets using an *80:20 ratio*.

Results

- Linear Regression outperformed the Random Forest Regressor, achieving an R^2 score of 0.6529 with lower MAE and RMSE values. This indicates that the model explains approximately 65% of the variation in house prices and captures the overall pricing trend effectively for this dataset.

- **Linear Regression:**
    - MAE: 970043.40
    - RMSE: 1324506.96
    - R^2 Score: 0.6529
- **Random Forest Regressor:**
    - MAE: 1010647.36
    - RMSE: 1391619.21
    - R^2 Score: 0.6169

Key Findings

Feature importance analysis showed that **house area** has the strongest influence on price, followed by **bathrooms, air conditioning, number of stories, parking availability, and bedrooms**. The distribution of house prices is positively skewed, with a small number of high-priced properties acting as outliers. Another notable observation is that amenities such as **air conditioning and preferred location contribute significantly to property value even when house size is similar**. The better performance of Linear Regression compared to Random Forest suggests that the relationship between the available housing features and price is predominantly linear in this dataset.

Recommendation

Real estate companies should prioritize **property area, number of bathrooms, parking availability, and amenities such as air conditioning** while estimating selling prices or evaluating new listings, as these features showed the strongest influence on house value in the trained model. Incorporating these variables into pricing strategies can help produce more consistent and data-driven property valuations.